MANIPAL
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MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

Machine Learning-Based Prediction And Recommendation Of Roommate Compatibility Using Lifestyle Preferences

Project Report

Applied Probability and Statistics Lab (AME 5152)

submitted to

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Abstract

Selecting a suitable roommate is a multi-attribute decision problem because individuals differ in their daily routines, lifestyle habits, financial requirements, and expectations about shared living spaces. This project investigates a machine learning-based approach for comparing roommate profiles and recommending potential matches. The supplied dataset contains 3,000 individual profiles and 37 columns, including an identifier and 36 descriptive attributes. The project workflow includes data inspection, preprocessing, exploratory analysis, pairwise feature construction, compatibility estimation, classification, and recommendation ranking. Logistic Regression and Random Forest classification results are evaluated using confusion matrices. Based on the supplied test results, Random Forest achieved 73.41% accuracy, while Logistic Regression achieved 62.92%. A paired bootstrap distribution is also included to study uncertainty in the accuracy difference. Since the original profile dataset does not contain observed compatibility outcomes, the origin and construction of the pairwise labels must be documented. The results should therefore be interpreted as performance against the labels used in the experiment, not as proof of real-world roommate satisfaction.

1. Problem Statement and Dataset

1.1 Problem Statement

Finding a suitable roommate can be challenging because individuals have different lifestyles, daily habits, personal preferences, and financial requirements. Differences in sleeping schedules, cleanliness expectations, noise tolerance, cooking habits, guest preferences, smoking preferences, and financial arrangements can create disagreements in shared accommodation. Traditional roommate selection often relies on limited information or informal discussions and may not systematically consider several factors together.

This project aims to develop a machine learning-based system that analyzes multiple lifestyle and preference attributes to estimate compatibility between two individuals and recommend suitable roommate candidates. The system compares profiles, considers essential living requirements, and ranks eligible candidates according to their preference compatibility.

1.2 Project Objectives

1. To inspect, clean, and statistically summarize the roommate lifestyle dataset.
2. To identify relevant personal, lifestyle, and financial attributes for roommate matching.
3. To construct pairwise features that represent similarities and potential conflicts between two profiles.
4. To implement classification approaches and compare their performance using appropriate evaluation metrics.
5. To develop a recommendation approach that ranks eligible roommate candidates and provides interpretable reasons for the match.
6. To apply probability and statistical techniques for sampling, analysis, model evaluation, and uncertainty estimation.

1.3 Dataset Description

The dataset used in this project is the uploaded file roommate_lifestyle_profiles_3000 (1).csv. It contains 3,000 rows and 37 columns. Each row represents one individual roommate profile. The

dataset includes the identifier User_ID and descriptive attributes covering personal details, financial preferences, lifestyle habits, and shared-living expectations.

Dataset property	**Value**
Dataset file	roommate_lifestyle_profiles_3000 (1).csv
Number of profiles	3,000
Number of columns	37
Identifier	User_ID
Descriptive attributes	36
Missing values	0
Exact duplicate rows	0
Original compatibility target	Not present in the profile CSV

1.4 Attribute Categories

Category	**Example attributes**	**Purpose in the project**
Personal profile	Age, Gender, Preferred_Roommate_Gender, City, Occupation	Profile description and eligibility filtering
Financial preferences	Monthly_Budget_INR, Budget_Flexibility, Financial_Payment_Style, Utilities_Split	Compare affordability and payment expectations
Lifestyle habits	Sleep_Schedule, Wake_Time, Cooking_Frequency, Smoking, Drinking, Pets, WFH_Frequency	Compare everyday routines and habits
Shared living	Room_Type, Kitchen_Sharing, Bathroom_Time, Cleanliness_Level, Noise_Tolerance,	Assess shared-space expectations

	Guest_Frequency, Party_Frequency	
Social and communication	Social_Level, Communication_Style, Conflict_Style, Roommate_Relationship	Understand social preferences and communication expectations
Study and work	Quiet_Hours_Importance, Schedule_Predictability, Study_Environment, Study_Hours_Daily	Compare work and study environments
Other preferences	Thermostat_Preference, Guest_Overnight_OK, Common_Space_Usage	Compare additional living preferences

1.5 Preliminary Dataset Statistics

The initial dataset inspection identified no missing values and no exact duplicate rows. The following descriptive statistics summarize selected numerical attributes.

Attribute	**Mean**	**Median**	**Minimum**	**Maximum**
Age	26.52	26.00	18	35
Monthly budget (INR)	14971.33	15000.00	5000	30000
Study hours daily	3.54	3.55	0	10

The mean monthly budget is approximately ₹14,971.33, and the median is ₹15,000. Age ranges from 18 to 35 years. Study hours range from 0 to 10 hours per day. These summaries describe the supplied dataset and do not establish that it represents the wider population of people seeking roommates.

An initial uniqueness check also found that Conflict_Style has only one distinct value in the supplied data (Calm Discussion). A constant feature cannot distinguish profiles and should be excluded from similarity calculations or model features.

1.6 Target Variable and Compatibility Definition

The original CSV contains individual profiles but no target column such as Compatibility_Score or Compatible/Not Compatible. A supervised classifier requires labelled examples. Therefore, the classification experiment must use a separate pair-level dataset in which each row represents a pair of individuals and the target indicates the compatibility label.

The report must state exactly how these pair labels were obtained. If labels were generated using a predefined rule or weighted score, the classifiers learn to reproduce that rule. Such an experiment can demonstrate a computational pipeline, but it cannot independently validate real-world compatibility. Stronger evidence would require compatibility ratings, post-match feedback, or other independently collected outcomes from actual roommate pairs.

2. Sampling and Experimental Methodology

2.1 Sampling Unit and Pair Formation

The original sampling unit is an individual profile. Compatibility, however, is a relationship between two profiles. Pairwise observations are therefore constructed by selecting two different individuals and deriving comparison features. A profile must not be paired with itself, and duplicate unordered pairs should be removed.

For n = 3,000 profiles, the number of unique unordered pairs is:

$N = n(n - 1) / 2 = 3{,}000 \times 2{,}999 / 2 = 4{,}498{,}500$ pairs

Generating all possible pairs may be unnecessary during initial development. A manageable random sample of pairs can be used for exploratory experiments, provided the sampling procedure is documented. If particular categories are rare, stratified sampling may be considered to ensure they are represented.

2.2 Train-Test Sampling and Data Leakage

The supplied confusion matrices each contain 267 test observations. The total test-set size is therefore 267 pair records for each model. The precise training-set size, validation procedure, split ratio, random seed, and whether the split was performed before or after pair generation should be added from the actual experiment code.

Pairwise rows are not necessarily independent because one person can appear in many different pairs. If pair rows are randomly divided between training and testing, the same individual may appear in both sets. This can cause information leakage and overstate generalization. A profile-level or group-aware split is recommended when the goal is to evaluate performance on previously unseen individuals.

2.3 Data Preprocessing

1. Load the CSV using Pandas and inspect its dimensions, column names, data types, and sample records.
2. Check missing values and duplicate rows. The initial inspection found zero missing values and zero exact duplicate rows.

3. Retain User_ID only to identify profiles and exclude it from predictive features.
4. Identify numerical, ordinal, and nominal categorical attributes.
5. Encode nominal categorical variables using an appropriate encoding method such as one-hot encoding.
6. Preserve the ordering of ordinal attributes where their scale has a meaningful order.
7. Scale numerical attributes when required by distance-based methods or models sensitive to feature scale.
8. Remove constant or uninformative features, including Conflict_Style in the supplied dataset.

2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the distributions, ranges, category frequencies, and potential relationships among attributes before model development. Suitable visualizations include histograms for Age and Monthly_Budget_INR, bar charts for Sleep_Schedule and Room_Type, frequency plots for Smoking and Pets, and boxplots for numerical attributes.

Correlation analysis may be used for appropriate numerical variables, while frequency tables and cross-tabulations can be used for categorical variables. EDA should be used to identify patterns and data-quality concerns, not to infer causal relationships.

2.5 Pairwise Feature Engineering

For each selected pair (A, B), pairwise features are created to express the relationship between the two profiles. Examples include the absolute difference in monthly budgets, whether sleep schedules match, the difference in cleanliness ratings, whether guest preferences are mutually acceptable, and whether both individuals can accept the proposed room arrangement.

Pairwise feature	**Example construction**	**Interpretation**
Budget difference	|Budget_A − Budget_B|	Smaller differences may indicate closer budget alignment
Sleep schedule agreement	1 if compatible, otherwise 0	Checks whether daily routines can coexist
Cleanliness difference	|Cleanliness_A − Cleanliness_B|	Measures difference in cleanliness expectations

Guest preference compatibility	Rule-based agreement indicator	Checks whether guest expectations are mutually acceptable
Social-level difference	|Social_Level_A − Social_Level_B|	Measures difference in social preferences
Room arrangement eligibility	Rule-based constraint	Checks whether both accept the proposed room type

Exact equality is not appropriate for every categorical attribute. For example, a person who accepts pets may be compatible with a person who owns a pet. Pairwise rules should therefore reflect the meaning of each attribute and distinguish acceptable differences from genuine conflicts.

2.6 Compatibility Scoring Baseline

A weighted similarity score can be used as an initial baseline when independently observed compatibility labels are unavailable. For a numerical feature j, normalized similarity may be defined as:

S□(A,B) = 1 − |xAj − xBj| / R□

where xAj and xBj are the values for individuals A and B, and Rj is the valid range of the feature. For categorical features, a compatibility function can assign a value according to whether the preferences match or are mutually acceptable.

C(A,B) = 100 × [Σ(w□ × S□(A,B)) / Σw□]

Here, wj is the importance assigned to feature j. The score is a designed compatibility index, not automatically a probability of successful cohabitation. Essential requirements, such as a maximum budget or a preferred roommate gender, should be applied as eligibility constraints rather than allowing a high score on unrelated attributes to override them.

2.7 Machine Learning Models

2.7.1 Logistic Regression

Logistic Regression is used as a baseline supervised classification algorithm. It estimates the probability of a pair belonging to the positive class using a logistic function applied to a linear combination of input features. In this project, the positive class is Compatible. A classification threshold is then used to assign the predicted class. Logistic Regression is relatively interpretable and provides a useful baseline for comparison.

2.7.2 Random Forest Classifier

Random Forest is an ensemble learning method that combines predictions from multiple decision trees. Each tree is trained using a sample of the training data and a subset of features. For classification, the trees vote for a class or contribute to an estimated class probability. Random Forest can model nonlinear relationships and interactions among pairwise lifestyle features. Its performance is evaluated using the same test observations as Logistic Regression.

2.7.3 Recommendation Stage

After a model or scoring mechanism evaluates candidate pairs, eligible candidates can be ranked from highest to lowest compatibility score or predicted probability. The recommendation interface should explain important matching factors and potential conflicts. A high score should be presented as a decision-support signal, not a guarantee that two people will live successfully together.

2.8 Evaluation Metrics

The confusion matrix provides True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The following measures are calculated with Compatible treated as the positive class.

Metric	Formula	Meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Proportion of all test pairs classified correctly

Precision	TP / (TP + FP)	Among pairs predicted Compatible, proportion labelled Compatible
Recall	TP / (TP + FN)	Proportion of labelled Compatible pairs identified
Specificity	TN / (TN + FP)	Proportion of labelled Not Compatible pairs identified
F1-score	2 × Precision × Recall / (Precision + Recall)	Harmonic mean of precision and recall

3. Observations, Decision, and Limitations

3.1 Random Forest Confusion Matrix

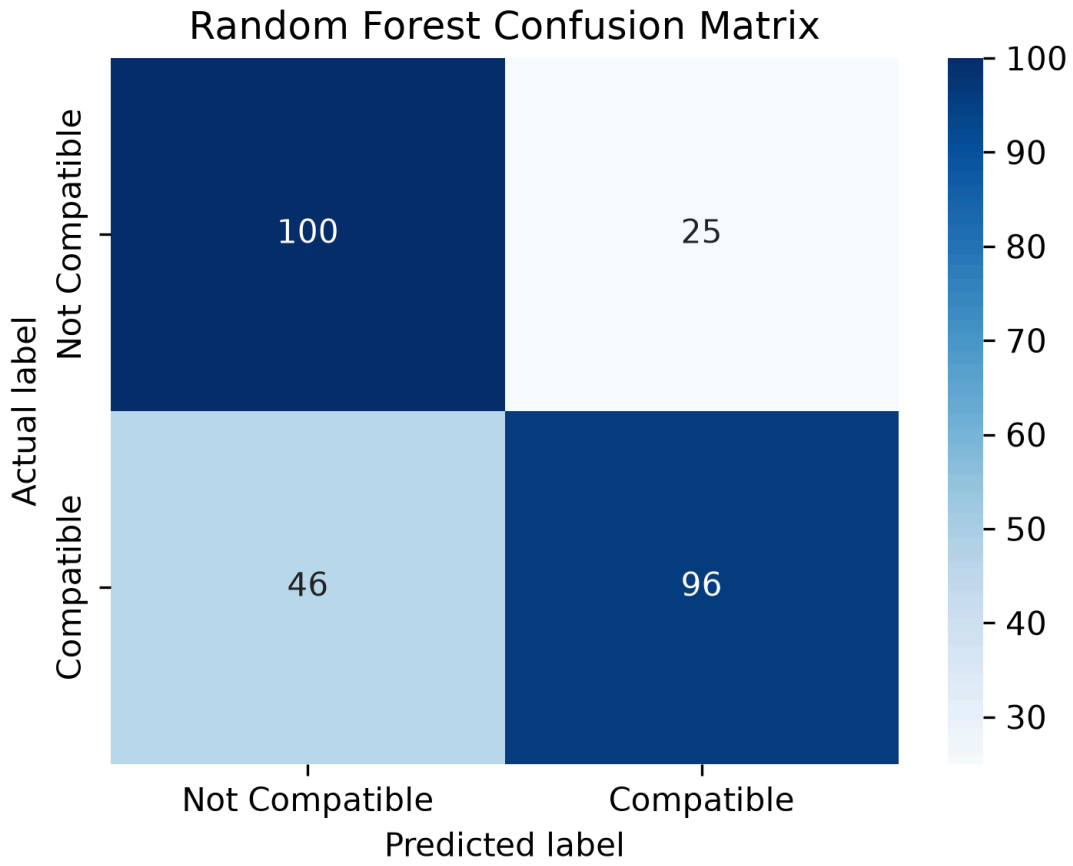


Figure 3.1: Confusion Matrix of the Random Forest Classifier

The Random Forest confusion matrix contains 100 true negatives, 25 false positives, 46 false negatives, and 96 true positives. Thus, 196 of the 267 test pairs were classified correctly.

Actual / Predicted	**Not Compatible**	**Compatible**
Not Compatible	100	25
Compatible	46	96

Accuracy = (100 + 96) / 267 = 73.41%. Precision for Compatible = 96 / (96 + 25) = 79.34%. Recall for Compatible = 96 / (96 + 46) = 67.61%. Specificity = 100 / (100 + 25) = 80.00%. The F1-score for Compatible is approximately 72.99%.

The model correctly identifies many incompatible pairs and has relatively high precision for the Compatible class. However, 46 labelled-compatible pairs are predicted as Not Compatible, indicating that some potentially suitable candidates are missed.

3.2 Logistic Regression Confusion Matrix

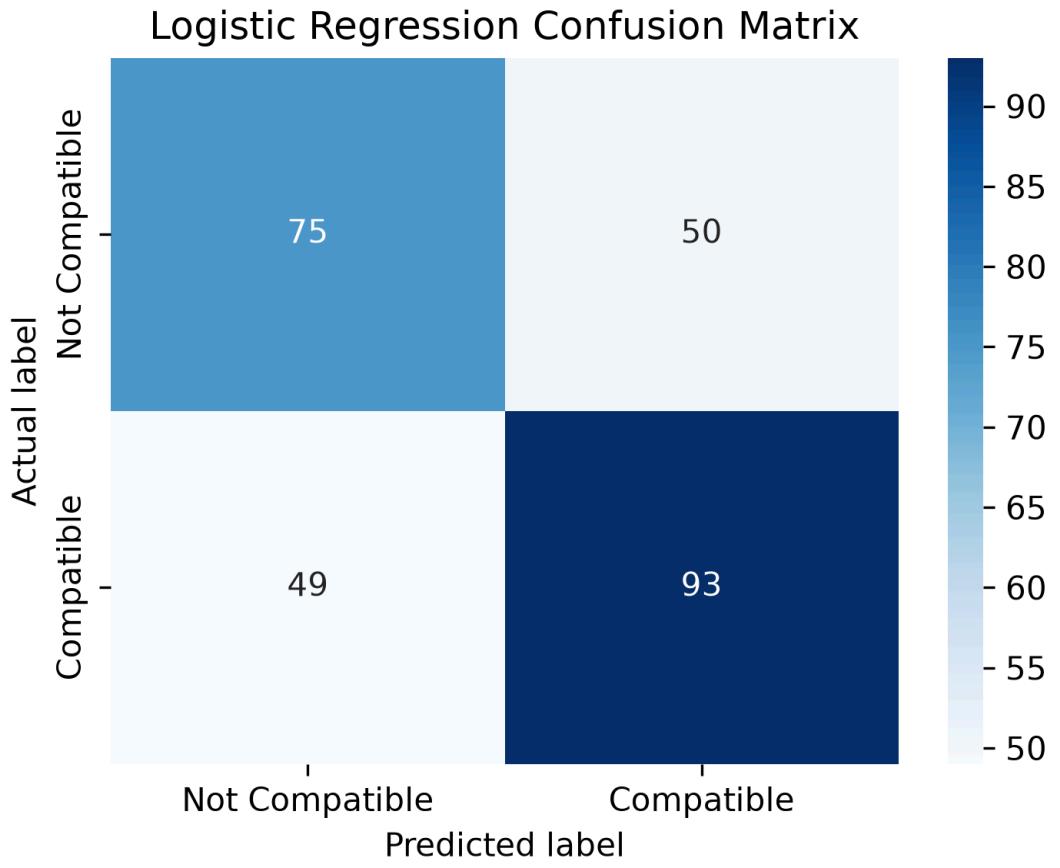


Figure 3.2: Confusion Matrix of the Logistic Regression Classifier

The Logistic Regression confusion matrix contains 75 true negatives, 50 false positives, 49 false negatives, and 93 true positives. Thus, 168 of the 267 test pairs were classified correctly.

Actual / Predicted	**Not Compatible**	**Compatible**
Not Compatible	75	50
Compatible	49	93

Accuracy = (75 + 93) / 267 = 62.92%. Precision for Compatible = 93 / (93 + 50) = 65.03%. Recall for Compatible = 93 / (93 + 49) = 65.49%. Specificity = 75 / (75 + 50) = 60.00%. The F1-score for Compatible is approximately 65.26%.

Logistic Regression identifies a similar number of compatible pairs correctly, but it produces more false positives and false negatives than Random Forest in this test set.

3.3 Comparison of Models

Evaluation metric	**Logistic Regression**	**Random Forest**
Accuracy	62.92%	73.41%
Precision (Compatible)	65.03%	79.34%
Recall (Compatible)	65.49%	67.61%
F1-score (Compatible)	65.26%	72.99%
Specificity	60.00%	80.00%

Random Forest outperforms Logistic Regression on all five reported metrics for the supplied test results. The accuracy difference is approximately 10.49 percentage points. Random Forest also has higher precision and specificity, indicating fewer false-positive compatibility predictions. Its recall is only moderately higher, so the model still misses some compatible pairs.

3.4 Observations and Decision

- The two confusion matrices each contain 267 test pair observations.
- Random Forest correctly classified 196 test pairs, while Logistic Regression correctly classified 168.
- Random Forest achieved 73.41% accuracy, compared with 62.92% for Logistic Regression.
- Random Forest achieved higher precision, recall, F1-score, and specificity on the supplied test set.
- Based on these results, Random Forest is selected as the better-performing model among the two evaluated classifiers for the current experiment.

3.5 Limitations

1. The original profile dataset does not contain observed compatibility labels. The source and construction of the pairwise target labels must be reported.

2. If labels are generated from a hand-designed score, model performance measures agreement with that score rather than real-world roommate success.
3. The representativeness and collection method of the profile dataset are not established by the CSV alone.
4. Preferences are subjective; different preferences do not always imply incompatibility, and matching preferences do not guarantee a successful living arrangement.
5. Manually selected feature weights can influence the baseline compatibility score.
6. Repeated appearances of the same individual across different pairs can create dependence and data leakage if splitting is performed only at the pair-row level.
7. Some attributes may be constant or have limited variation. A constant feature such as Conflict_Style cannot help discriminate between profiles.
8. Real-life compatibility may change over time as routines, budgets, and personal circumstances change.

4. Application of Probability and Statistics

4.1 Descriptive Statistics

Descriptive statistics summarize numerical attributes and help explain the characteristics of the available profiles. The mean is the arithmetic average, the median is the middle observation after sorting, and standard deviation measures dispersion around the mean.

Mean = Σx_i / n

Variance = $\Sigma(x_i - \bar{x})^2$ / n (population form)

Standard deviation = √Variance

For example, the dataset's mean monthly budget is ₹14,971.33 and its median is ₹15,000. These values describe the budget distribution in the supplied data. The range is ₹5,000 to ₹30,000.

4.2 Frequency and Probability Distributions

For a categorical attribute, the relative frequency of a category can be used as an empirical probability estimate:

P(X = category) = Count of profiles in category / Total profiles

For example, the frequency distribution of Sleep_Schedule can show the proportion of profiles in each sleep category. Similar calculations can be performed for Smoking, Pets, Room_Type, Guest_Frequency, and other categorical attributes. These are empirical proportions within the supplied dataset, not necessarily population probabilities.

4.3 Conditional Probability

Conditional probability measures the probability of an event given that another event has occurred:

P(A | B) = P(A ∩ B) / P(B), provided P(B) > 0

In this project, conditional probability can be used to examine questions such as the proportion of profiles preferring a private room among people in a particular occupation, or the proportion of individuals with a particular sleep schedule among those who work from home frequently. These analyses reveal associations and should not be interpreted as causal effects.

4.4 Sampling and Sampling Probability

The 3,000 profiles produce 4,498,500 possible unique unordered pairs. Sampling a subset of pairs can reduce computation. If m pairs are selected from N eligible pairs, the sampling fraction is m/N. Under simple random sampling without replacement, each eligible pair has an equal chance of selection. Stratified sampling can be considered if particular preference groups are underrepresented.

The test set shown in the supplied confusion matrices contains 267 pair records. The report should state the actual sampling method, split ratio, and random seed used to create this test set. If pairs share profiles, a group-aware design is preferable.

4.5 Correlation and Association Analysis

Correlation can be used to study associations between numerical attributes such as Age and Monthly_Budget_INR, or Study_Hours_Daily and Quiet_Hours_Importance. Pearson's correlation coefficient measures linear association:

$r = Cov(X,Y) / (\sigma X \, \sigma Y)$

The coefficient ranges from −1 to +1. Positive values indicate positive linear association, negative values indicate negative linear association, and values near zero indicate weak linear association. Spearman rank correlation may be more suitable for ordinal variables or monotonic relationships. Correlation does not imply causation.

4.6 Confusion Matrix as a Statistical Evaluation Tool

The confusion matrices summarize the observed outcomes of classification on the test set. Accuracy, precision, recall, specificity, and F1-score are calculated from the four confusion-matrix counts. Precision and recall are particularly relevant because a false-positive recommendation and a missed compatible candidate have different practical consequences.

Metric	Logistic Regression	Random Forest
True negatives	75	100
False positives	50	25
False negatives	49	46
True positives	93	96

4.7 Paired Bootstrap Analysis

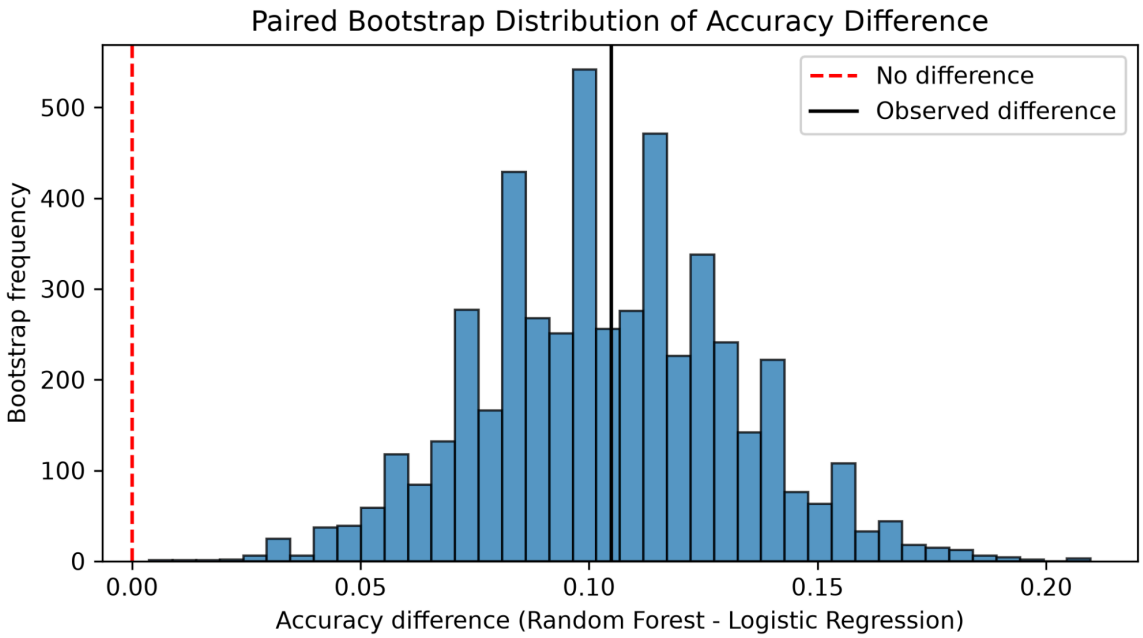


Figure 4.1: Paired Bootstrap Distribution of Accuracy Difference (Random Forest − Logistic Regression)

Bootstrap resampling is a statistical technique used to estimate the sampling variability of a statistic. In a paired comparison, the same resampled test observations must be used to evaluate both classifiers in each iteration. This preserves the pairing between model predictions on the same cases.

For each bootstrap sample, calculate the accuracy of Random Forest and Logistic Regression, then calculate their difference:

D_β = Accuracy(RF)$_\beta$ − Accuracy(LR)$_\beta$

The observed accuracy difference from the confusion matrices is:

0.7341 − 0.6292 = 0.1049, or approximately 10.49 percentage points

In the supplied histogram, the red dashed line at zero represents no accuracy difference, and the black vertical line represents the observed difference. The displayed distribution is concentrated on the positive side of zero, suggesting that Random Forest tends to have higher accuracy in the bootstrap resamples shown.

The histogram is a visual summary only. A formal statistical conclusion requires the numerical bootstrap results. A percentile-based 95% confidence interval can be calculated from the 2.5th and 97.5th percentiles of the bootstrap differences. A bootstrap p-value may also be calculated using a

pre-specified testing procedure. The exact interval, p-value, and number of iterations should be copied from the code output; they must not be estimated by visual inspection.

Because roommate pairs can share individuals, ordinary row-wise bootstrap resampling may not fully account for dependence among pairs. A profile-level or other cluster-aware bootstrap may be more appropriate if the aim is to generalize to new individuals.

4.8 Overall Role of Probability and Statistics

Project stage	**Concept**	**Use**
Dataset understanding	Mean, median, range, standard deviation	Summarize numerical profile attributes
EDA	Frequency and relative frequency	Describe lifestyle-category distributions
Preference analysis	Conditional probability	Examine relationships between categories
Pair construction	Random and stratified sampling	Select manageable and representative pairs
Exploratory analysis	Correlation and association	Study relationships between variables
Classification	Confusion matrix and derived metrics	Quantify prediction performance
Model comparison	Paired bootstrap distribution	Estimate uncertainty in accuracy difference

5. Conclusion

This project investigates a machine learning-based approach for predicting and recommending potential roommate matches using lifestyle preferences. The supplied dataset contains 3,000 individual profiles and a combination of numerical and categorical attributes related to personal information, finances, daily routines, and shared-living expectations. The methodology converts individual profiles into pairwise comparisons and evaluates classification approaches.

Based on the supplied confusion matrices, Random Forest achieved 73.41% accuracy, while Logistic Regression achieved 62.92%. Random Forest also achieved higher precision, recall, F1-score, and specificity on the evaluated test set. The paired bootstrap distribution provides a visual statistical comparison of the accuracy difference; a numerical confidence interval and appropriate inferential result should be added before claiming statistical significance.

Probability and statistics contribute throughout the project through descriptive summaries, empirical probability distributions, conditional probability, sampling, correlation analysis, classification metrics, and bootstrap resampling. The final system can provide ranked candidate recommendations and explanations of matching preferences. However, its real-world predictive validity depends on obtaining reliable compatibility labels and evaluating the system without profile-level leakage.